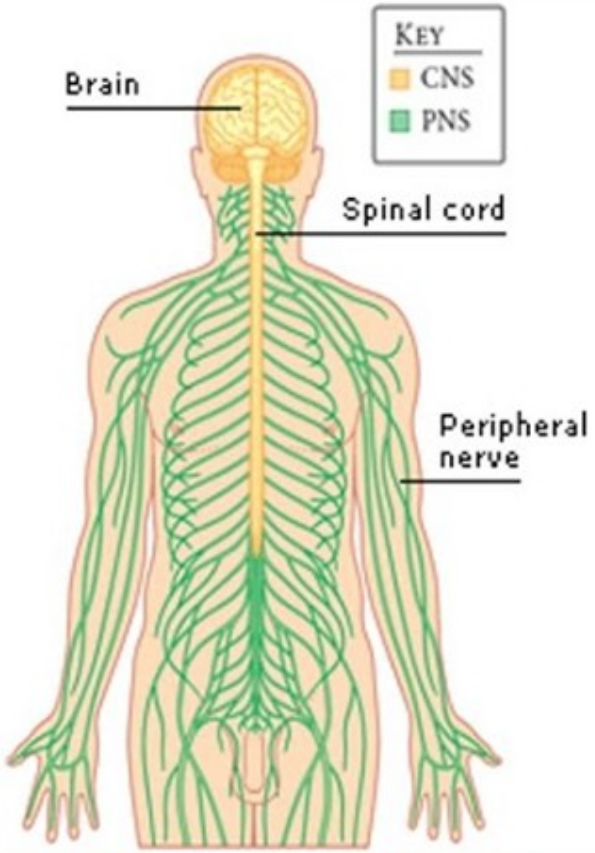
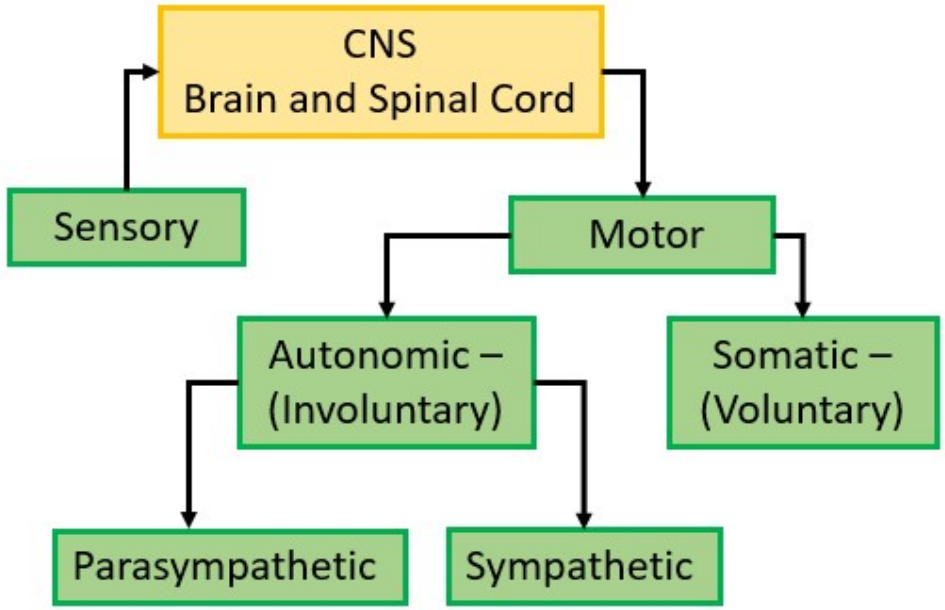
Divisions of the Nervous System: Peripheral Nervous System (PNS) – Motor Reflexes

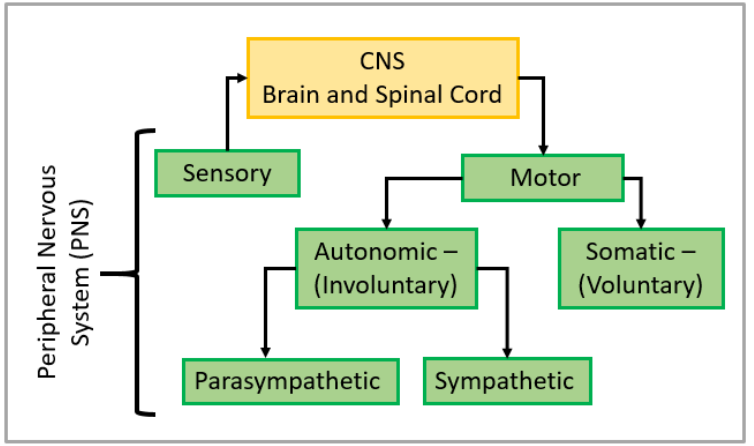


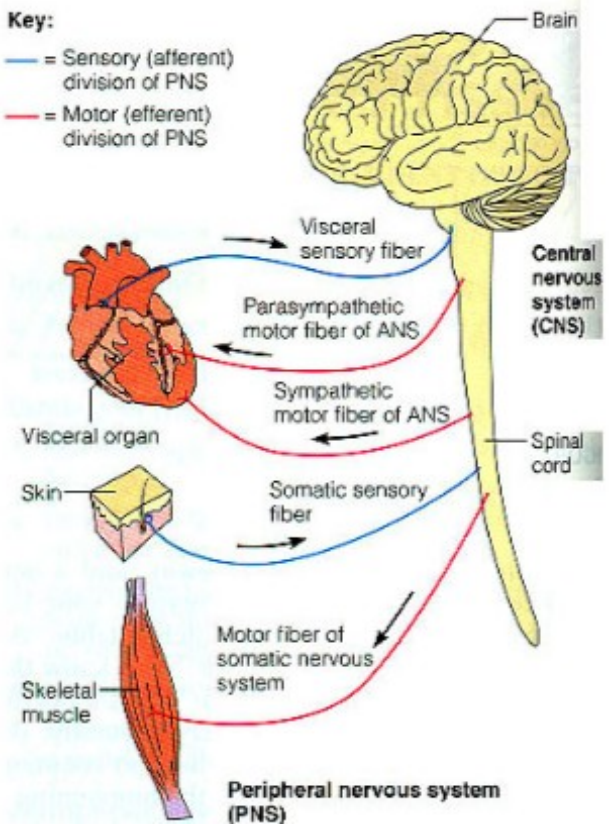


The nervous system is highly organised

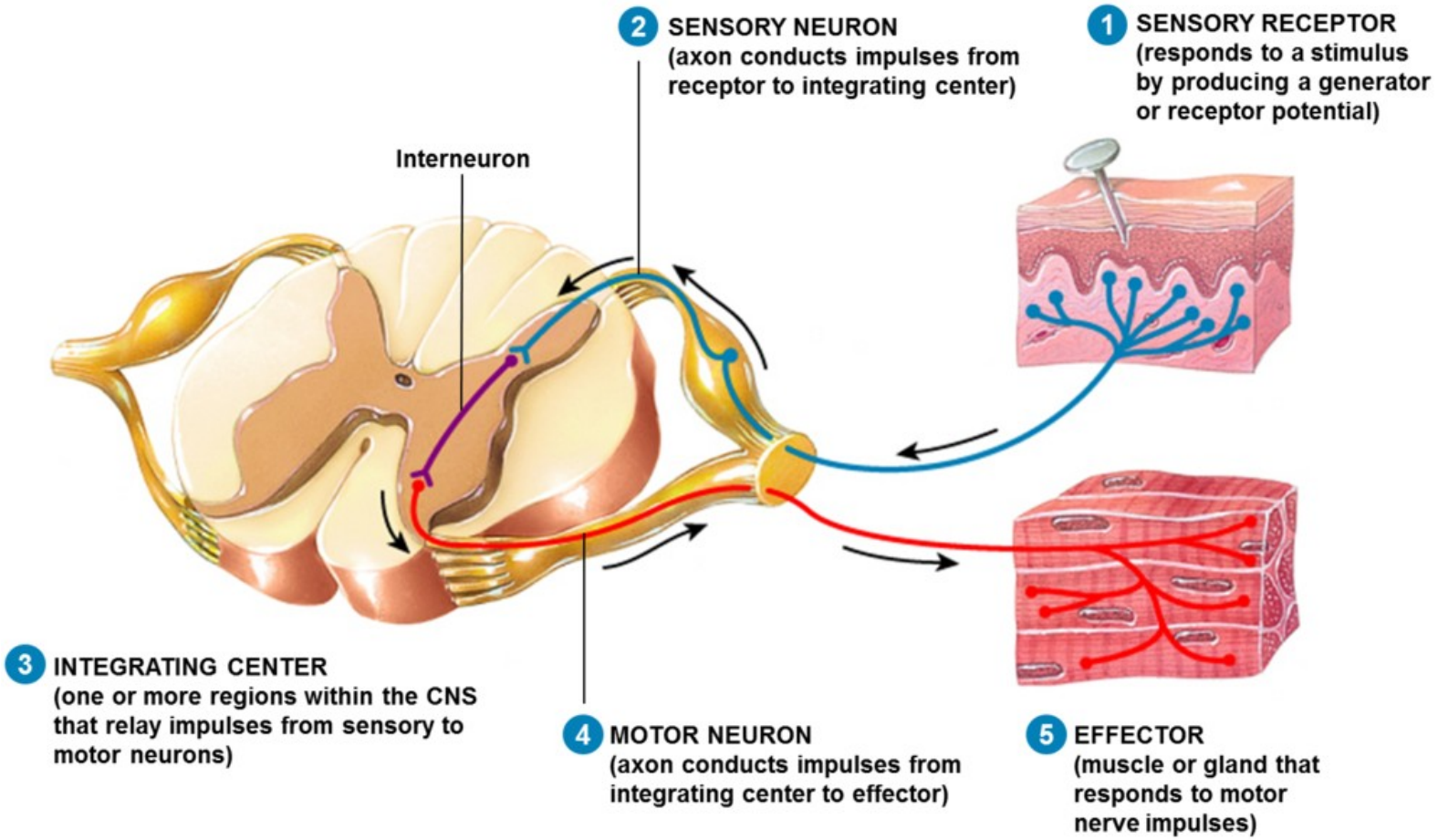
Efferent (motor) Division

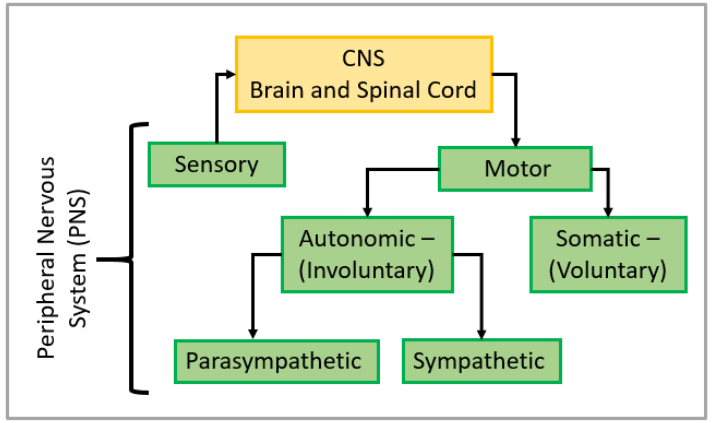
- Takes signals ***from CNS*** to body organs and systems to respond
- Somatic:
  - CNS to skeletal muscles
  - Contraction, muscle tone and movement
  - Some voluntary control
- Autonomic:
  - CNS to smooth muscle in body organs, and glands
  - Involuntary – not under conscious control
  - 2 divisions:
    - Sympathetic
    - Parasympathetic



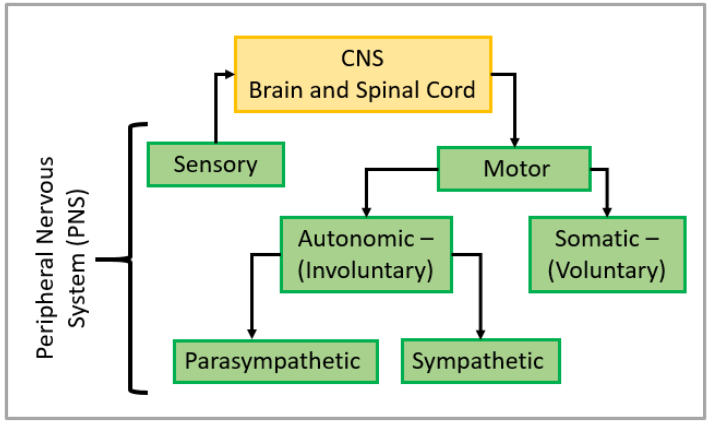


Learning Aim: List the parts and functions of the motor division of the PNS

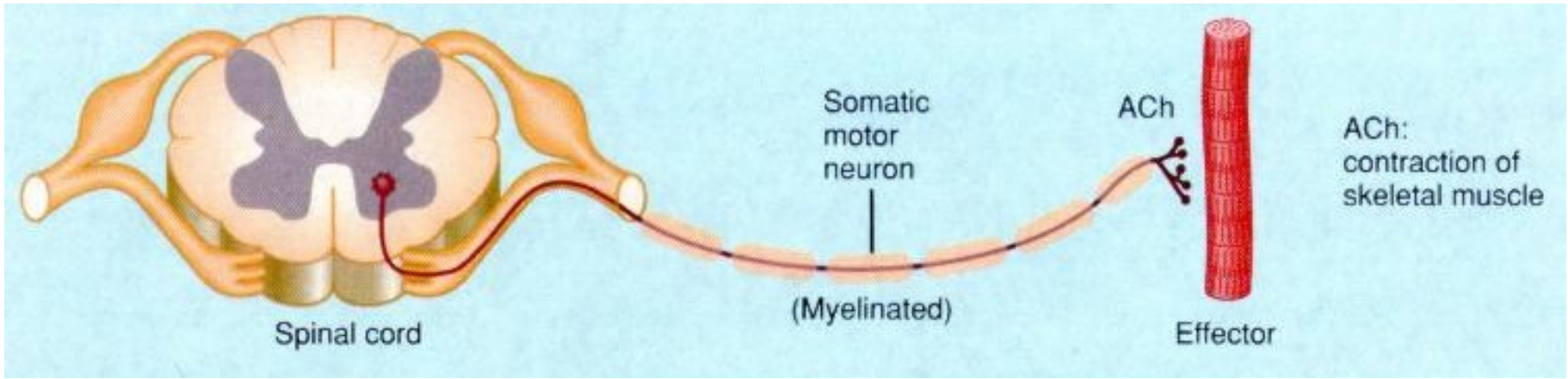

2 SENSORY NEURON
(axon conducts impulses from receptor to integrating center)
1 SENSORY RECEPTOR
(responds to a stimulus by producing a generator or receptor potential)
Interneuron
3 INTEGRATING CENTER
(one or more regions within the CNS that relay impulses from sensory to motor neurons)
4 MOTOR NEURON
(axon conducts impulses from integrating center to effector)
5 EFFECTOR
(muscle or gland that responds to motor nerve impulses)

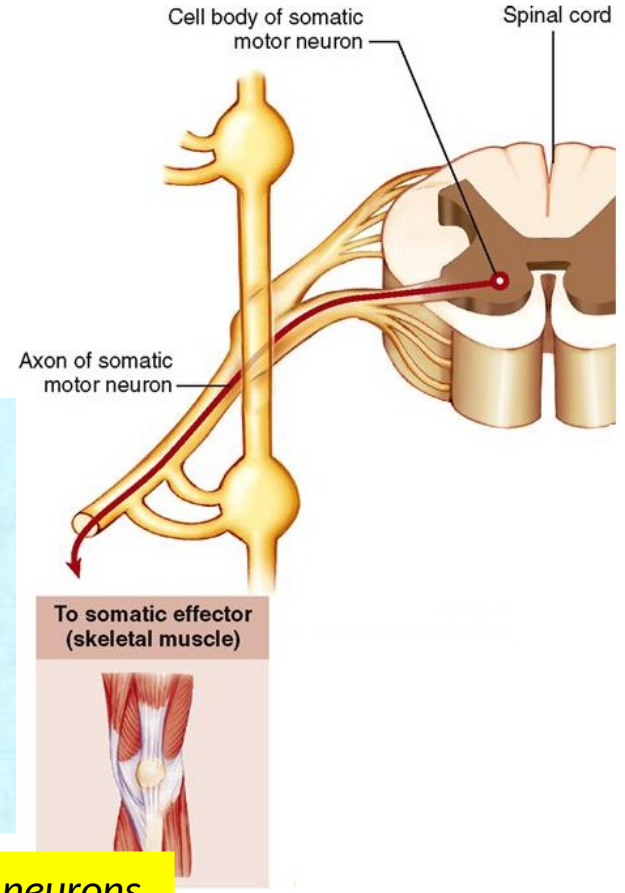

CNS
Brain and Spinal Cord
Peripheral Nervous System (PNS)
Sensory
Motor
Autonomic – (Involuntary)
Somatic – (Voluntary)
Parasympathetic
Sympathetic

Somatic Division of Motor PNS



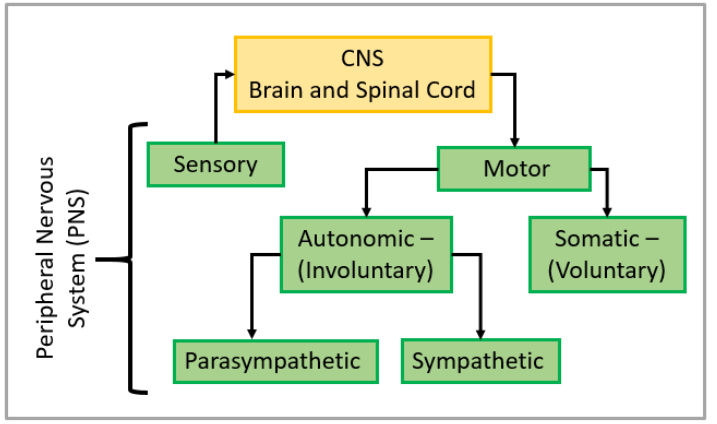
- ***CNS to skeletal muscles***
  - Acetylcholine (ACh) release stimulates muscle contraction
  - Under mostly conscious control
- Cell bodies are in CNS (eg spinal cord)
- Axons extend from CNS all the way to muscles
  - No intermediate synapses or ganglia



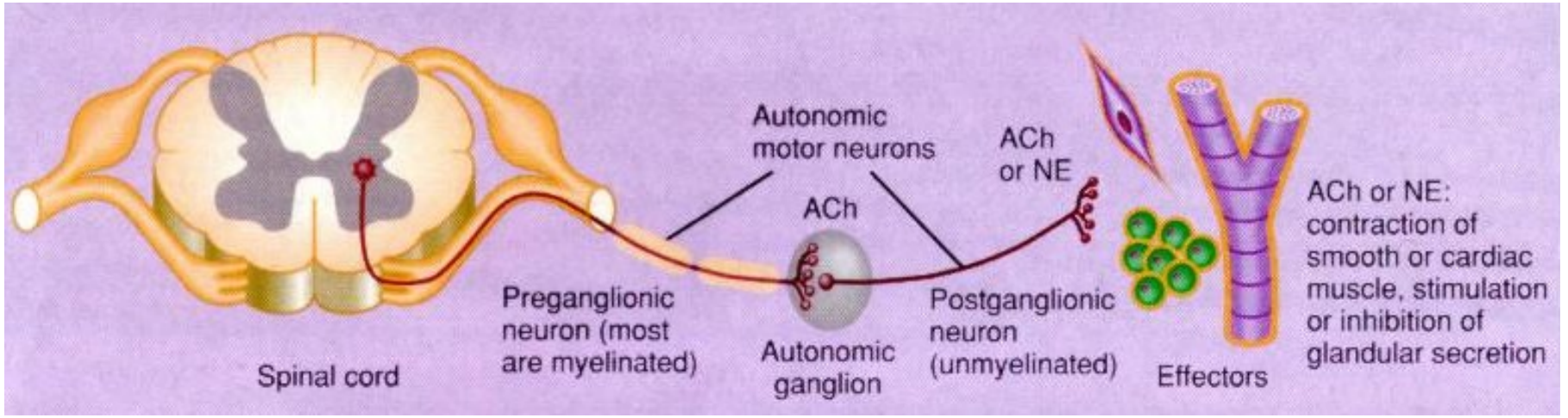


Learning Aim: Describe the pathway, structure and function of somatic motor neurons

Autonomic Division of Motor PNS



- Signals from CNS to smooth muscle inside body organs
- Mostly involuntary
- Two divisions: Sympathetic and Parasympathetic
- Involved in adjustment of internal environment: homeostasis
- Motor neuron from CNS to autonomic ganglion, then another to effector tissue.



Learning Aim: Describe the pathway, structure and function of autonomic motor neurons

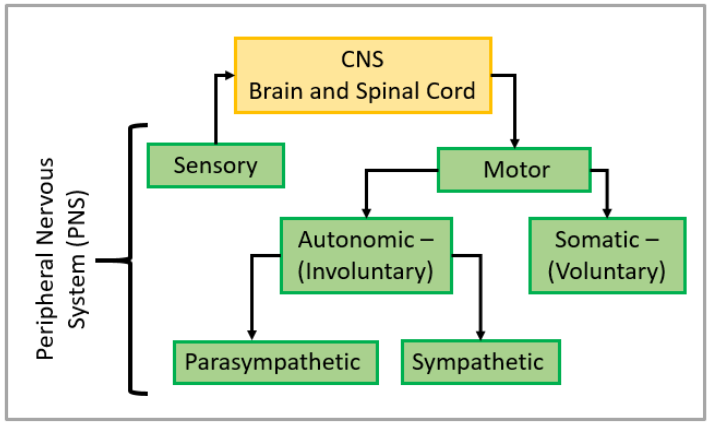


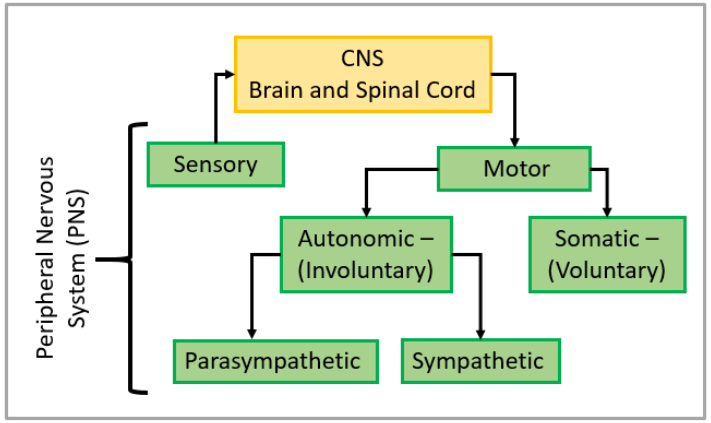
Table 4.1 A comparison between the autonomic and somatic divisions of the peripheral nervous system

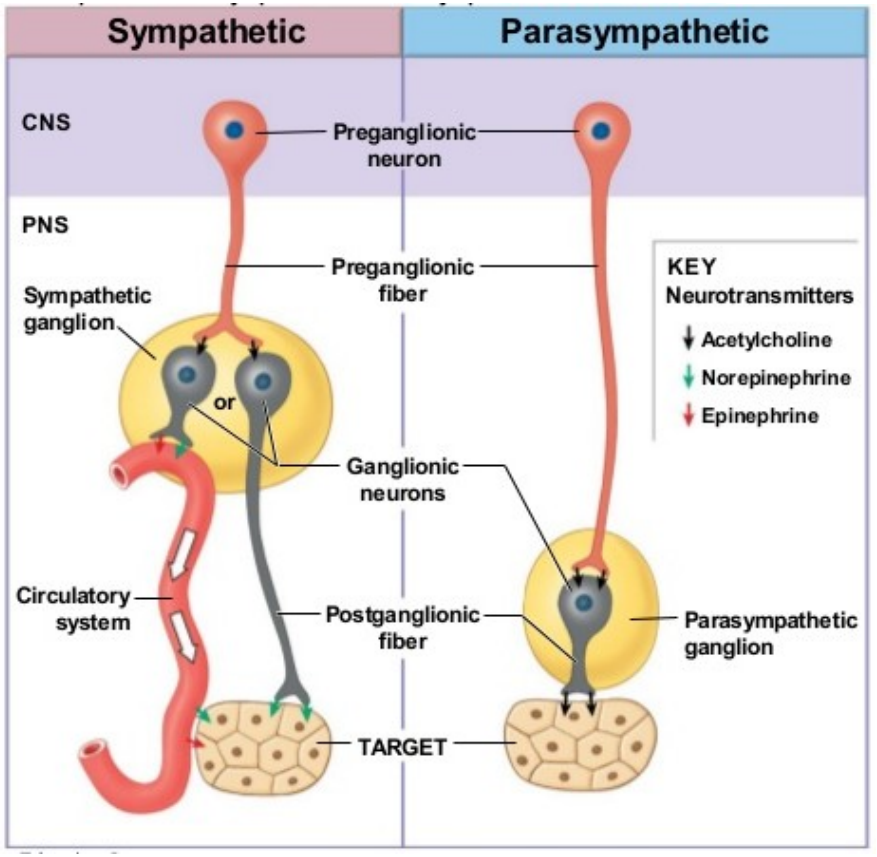
Characteristic	Autonomic division	Somatic division
Effectors	Heart muscle, involuntary muscle, glands	Skeletal (voluntary) muscles
General function	Adjustment of the internal environment (homeostasis)	Response to the external environment
Efferent (outward) pathways	Two nerve fibres from the CNS to the effector with a synapse in a ganglion	One nerve fibre from the CNS to the effector; no synapse and no ganglion
Neurotransmitter at effector	Acetylcholine or noradrenaline	Acetylcholine
Control	Usually involuntary	Usually voluntary
Nerves to target organ	Two sets – sympathetic and parasympathetic	One set
Effect on target organ	Excitation or inhibition	Always excitation

Autonomic Division of Motor PNS

- Sympathetic:
  - “fight or flight”
  - Norepinephrine (noradrenaline) released at tissues
  - Ganglia closer to CNS

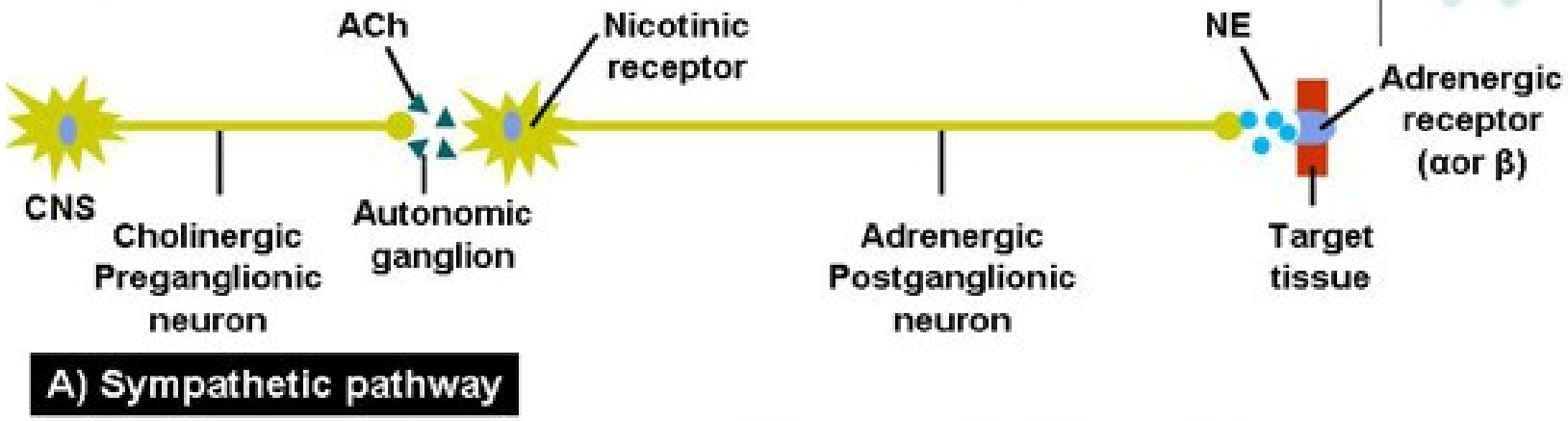
- Parasympathetic
  - “rest and digest”
  - Acetylcholine (ACh) released at tissues
  - Ganglia further from CNS



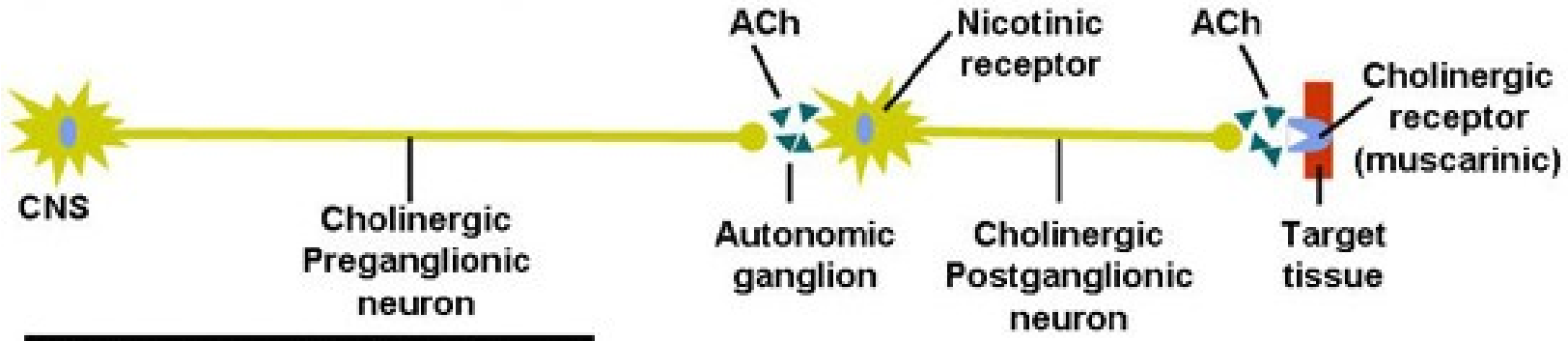


Learning Aim: Compare and contrast the pathway, structure and function of sympathetic and parasympathetic autonomic motor neurons

ANS Neurotransmitters

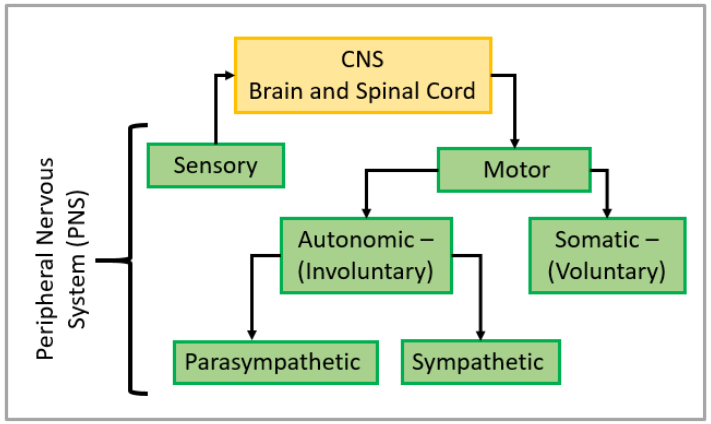


A) Sympathetic pathway



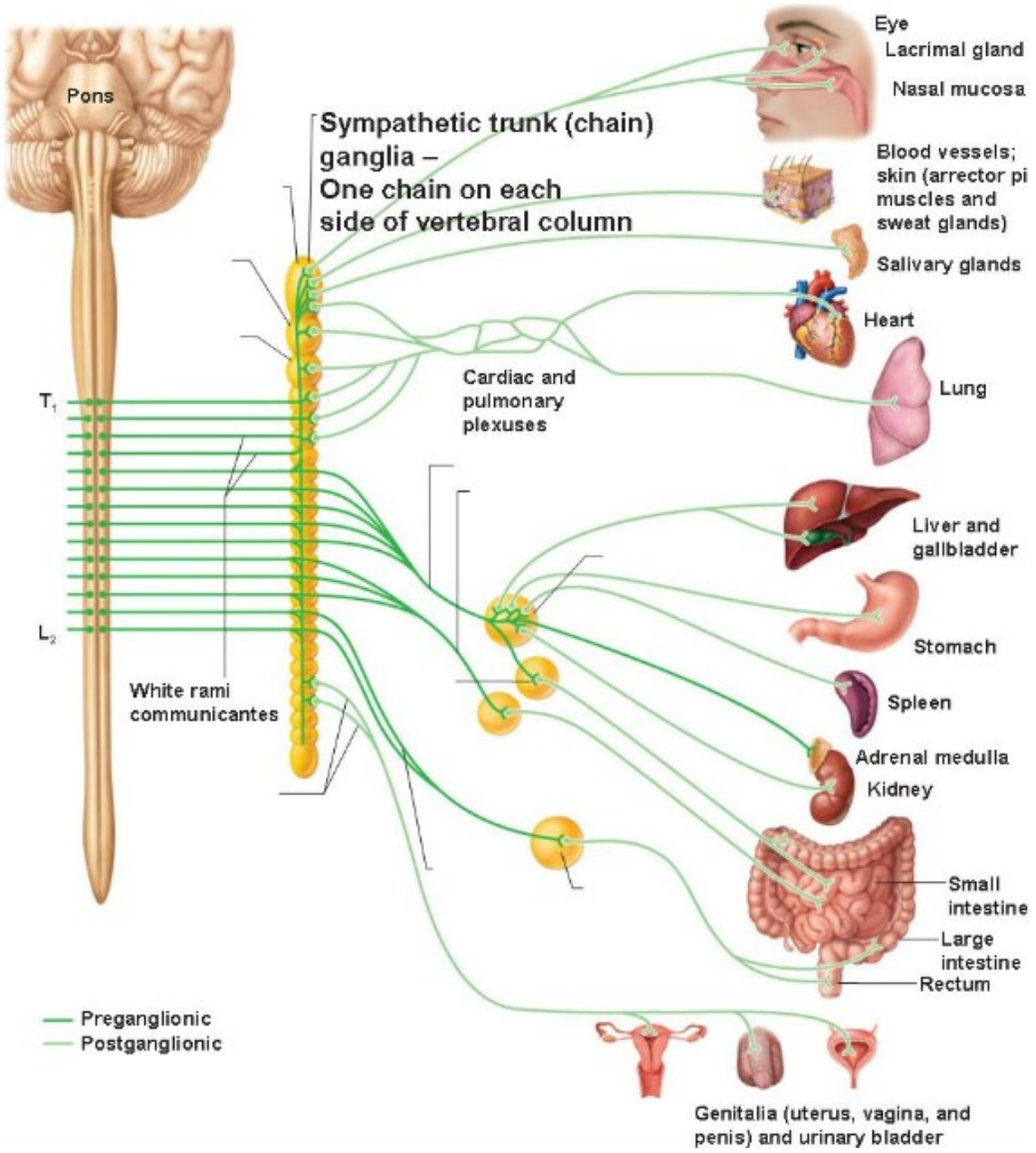
B) Parasympathetic pathway

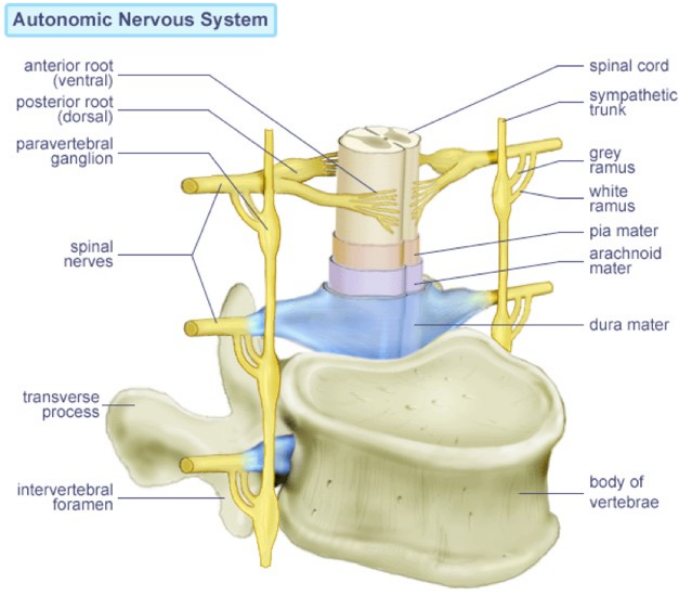
ACh = Acetylcholine
NE = Norepinephrine

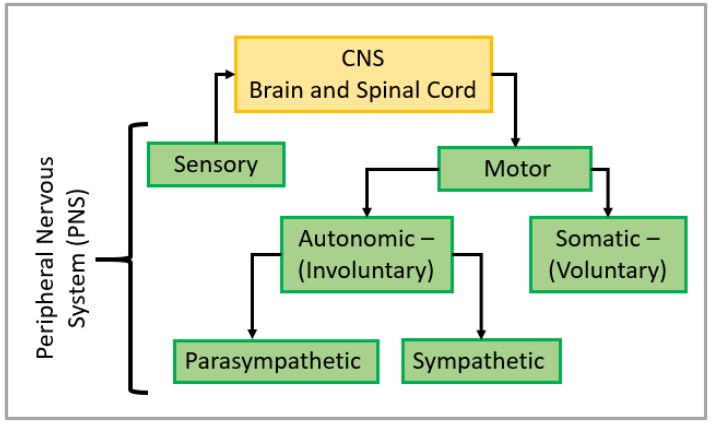


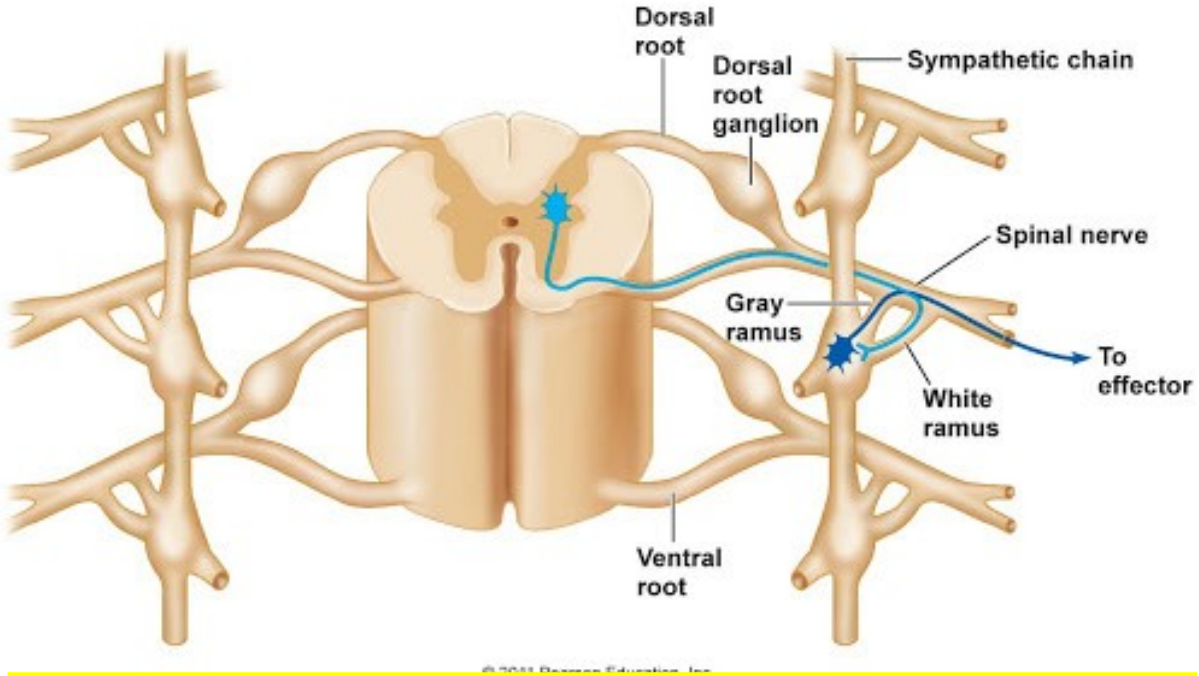
Learning Aim: Describe the pathway, structure and function of somatic motor neurons

Organisation of the Sympathetic NS



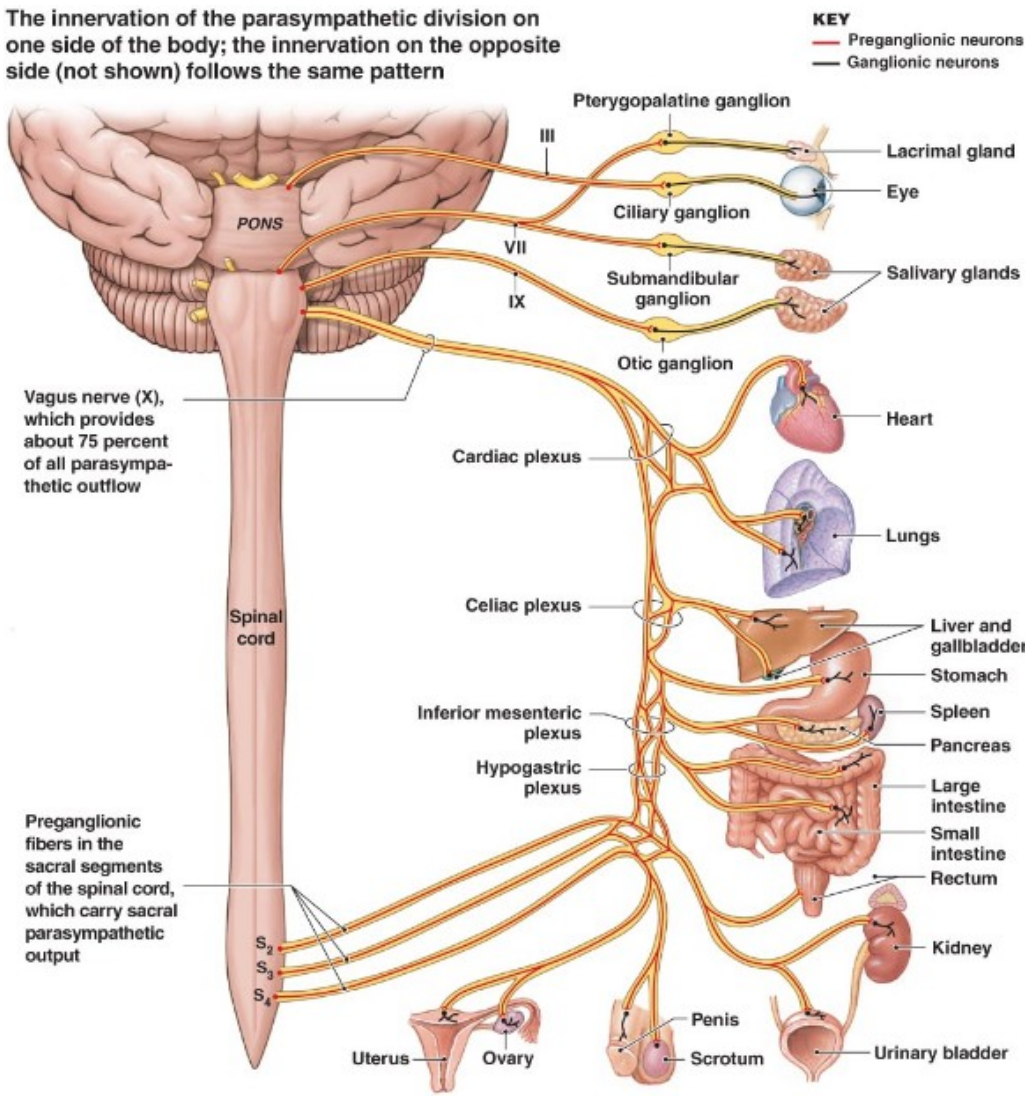


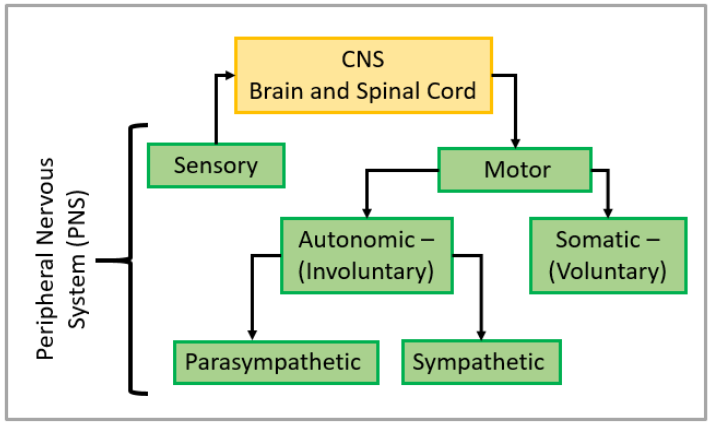




Learning Aim: Describe the broad structure of the sympathetic NS

Organisation of the Parasympathetic NS



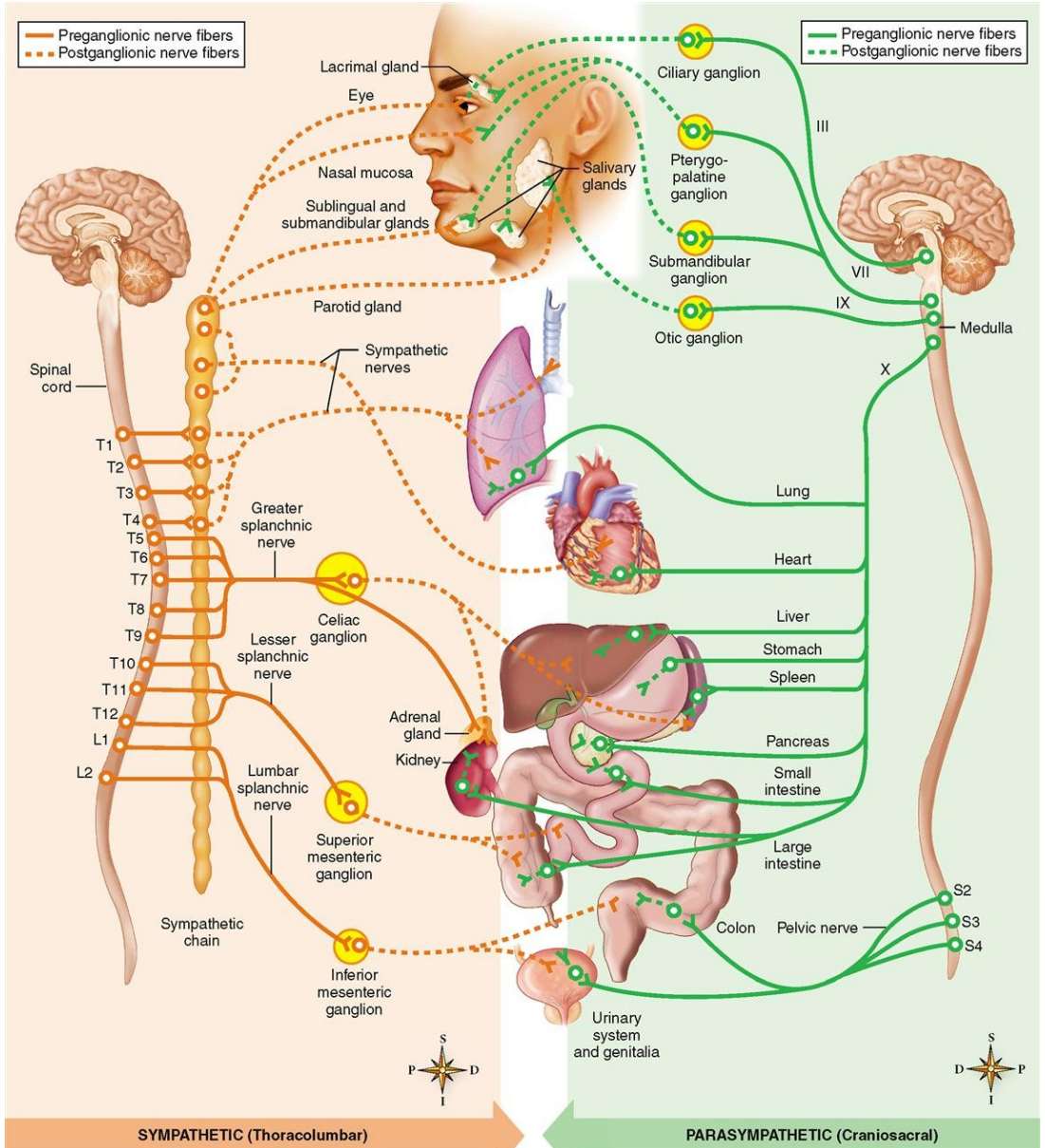


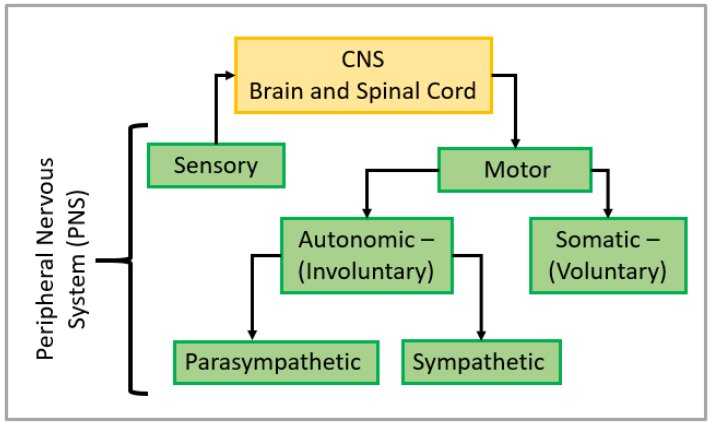
Parasympathetic Neurons emerge from the brain and from the base of the spinal cord, not from the ventral roots of most segments of the spinal cord.

(Sympathetic and motor fibers emerge from the ventral roots of most spinal segments.)

Learning Aim: Describe the broad structure of the parasympathetic NS

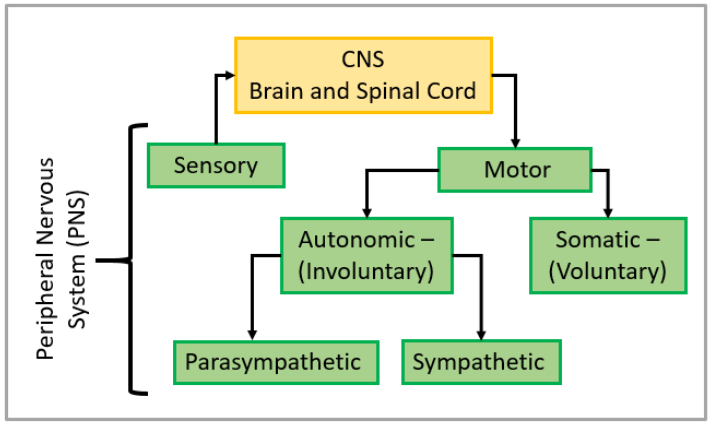
Visceral Organs and Blood vessels are stimulated by both Sympathetic and Parasympathetic neurons.





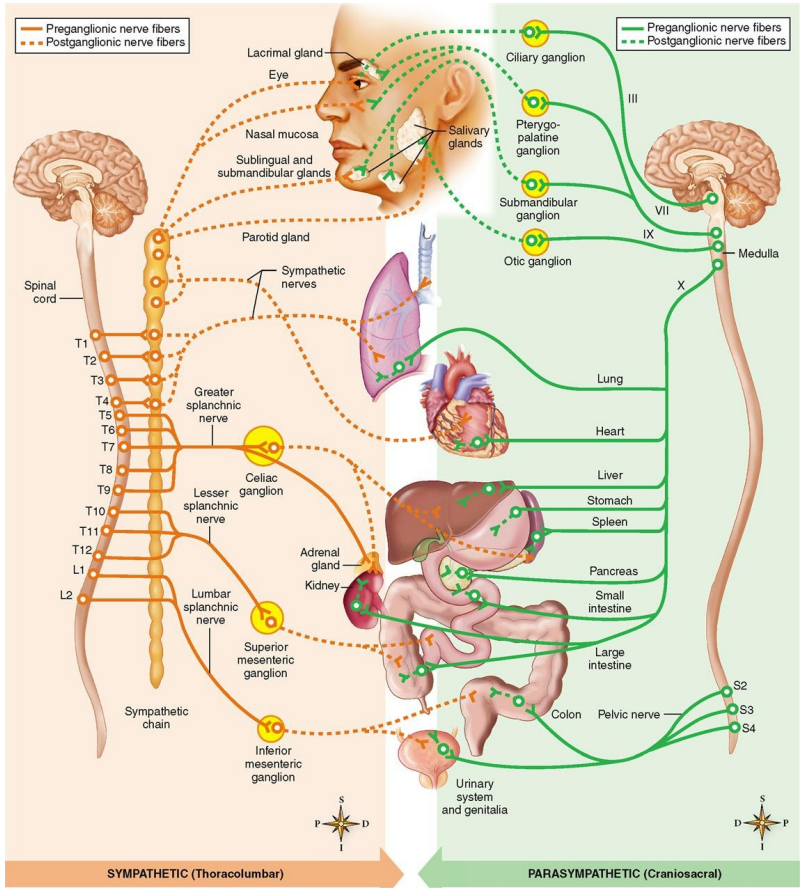
Structure	Effect of sympathetic stimulation	Effect of parasympathetic stimulation
Heart	Increases rate and strength of contraction	Decreases rate and strength of contraction
Lungs	Dilates bronchioles (fine air passages in the lungs)	Constricts bronchioles
Stomach, intestines	Decreases movement	Increases movement
Liver	Increases breakdown of glycogen and release of glucose	Increases uptake of glucose and synthesis of glycogen
Iris of the eye	Dilates pupil	Constricts pupil
Sweat glands	Increases sweat secretion	No effect
Salivary glands	Decreases secretion of saliva	Increases secretion of saliva
Blood vessels of:		
skin	Constricts vessels	Little effect
skeletal muscle	Dilates vessels	No effect
internal organs	Constricts vessels (except in heart and lung)	Little effect
Urinary bladder	Relaxes muscles of wall	Constricts muscles of wall
Adrenal medulla	Stimulates hormone secretion	No effect

- Sympathetic and Parasympathetic systems usually balance each other to maintain homeostasis
- each target tissue has both sympathetic and parasympathetic neuron supply
- in extreme circumstances, the sympathetic dominates as a survival strategy:



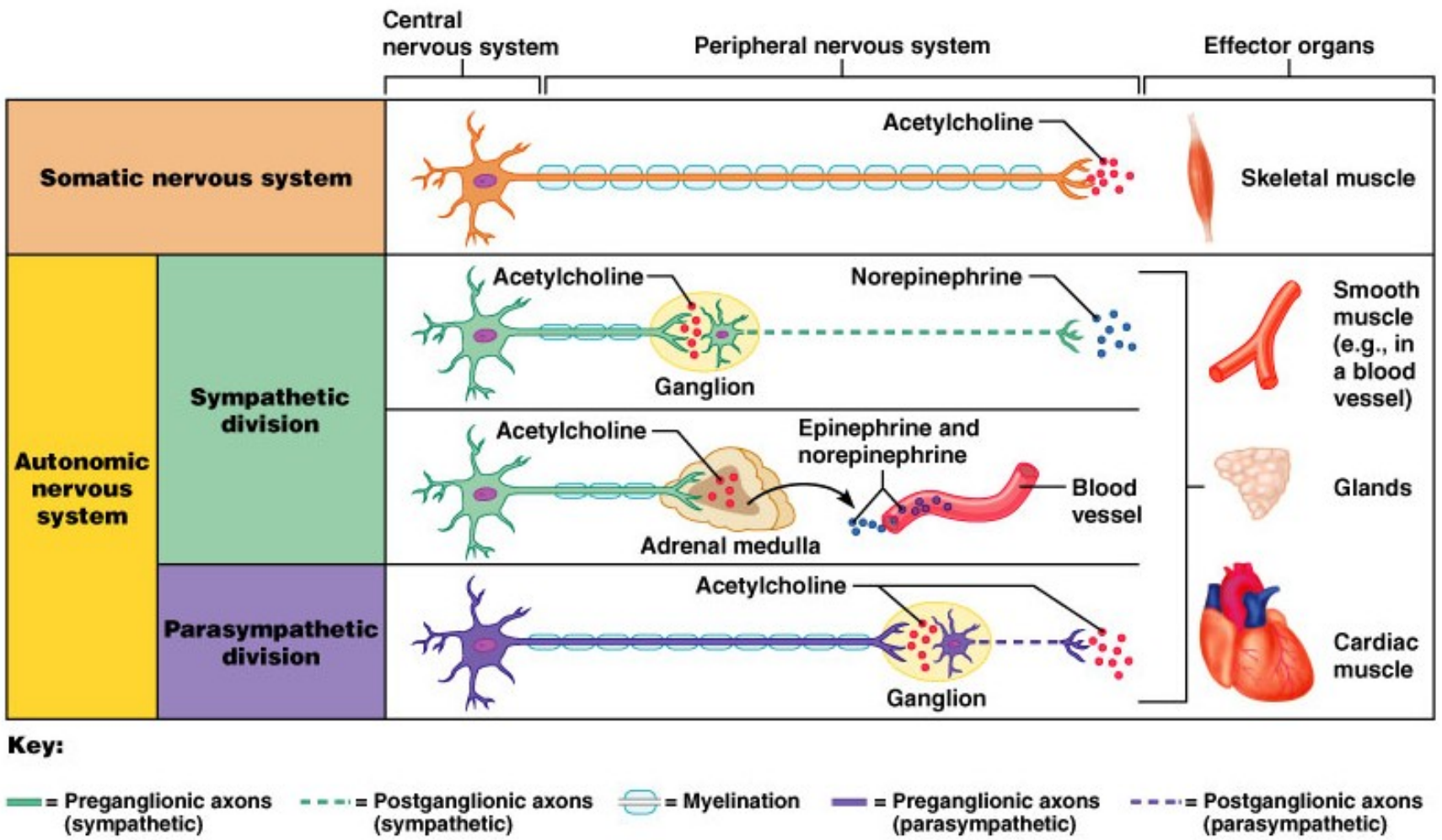
“Fight or Flight”

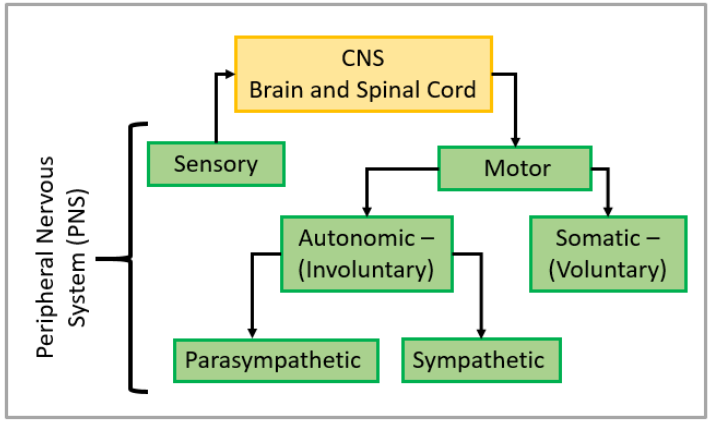
- Body prepares for increased activity (eg metabolism, O_2 needs)
  - Increase rate and force of heart contraction → increased bp
  - Blood vessels at muscles, heart and liver dilate
  - Blood vessels at skin, digestive system, kidney constrict
  - Airways dilate
  - Sweat glands secrete more
  - Adrenal medulla stimulated to produce adrenalin/noradrenalin to intensify these responses.



Learning Aim: explain the elements of the Fight or Flight response

PNS Motor Neurons, synapses and neurotransmitters





Learning Aim: Describe the synapses and neurotransmitters involved in the different parts of the motor division of the PNS

Reflexes



The nervous system is highly organised

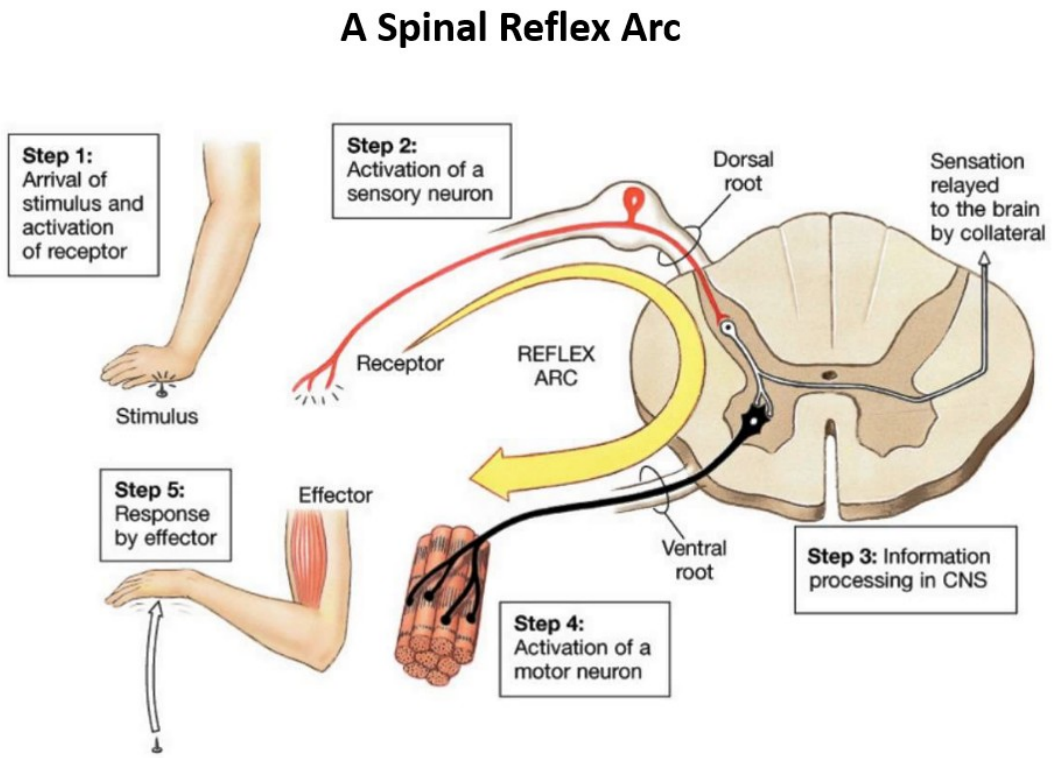
Reflexes

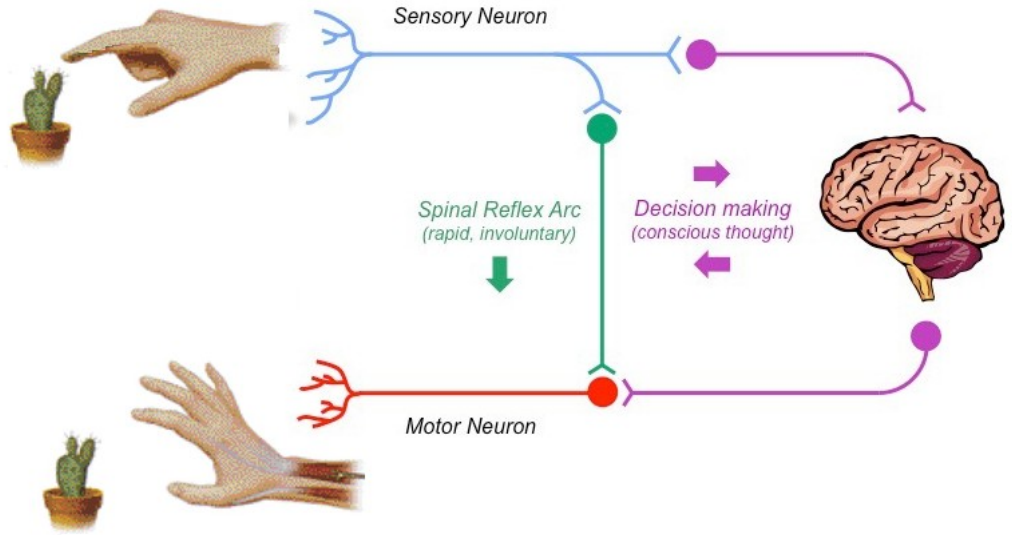
- Rapid, involuntary response to change.
- 4 properties:
  - Stimulus required
  - Involuntary
  - Rapid
  - Stereotyped (occurs same way each time)

- May involve:
  - Unconscious parts of the brain (sometimes)
  - Spinal Cord (usually)

- Function to
  - Protect body from injury
  - Maintain homeostasis

Learning Aim: Define reflexes and list the four properties of reflexes





Components of a Reflex Arc

Receptor:	Control Centre	Effector:
• Provides stimulus, eg • Pain • Change in homeostatic balance	• Brain • Spinal Cord	• Tissue/organ/system that responds eg: • Skeletal muscle • Smooth muscle • Gland

Has the following components:

A receptor

- End of sensory neuron, or a specialised cell that transmits to a sensory neuron

A Sensory neuron

- Carries impulses to CNS

At least one synapse

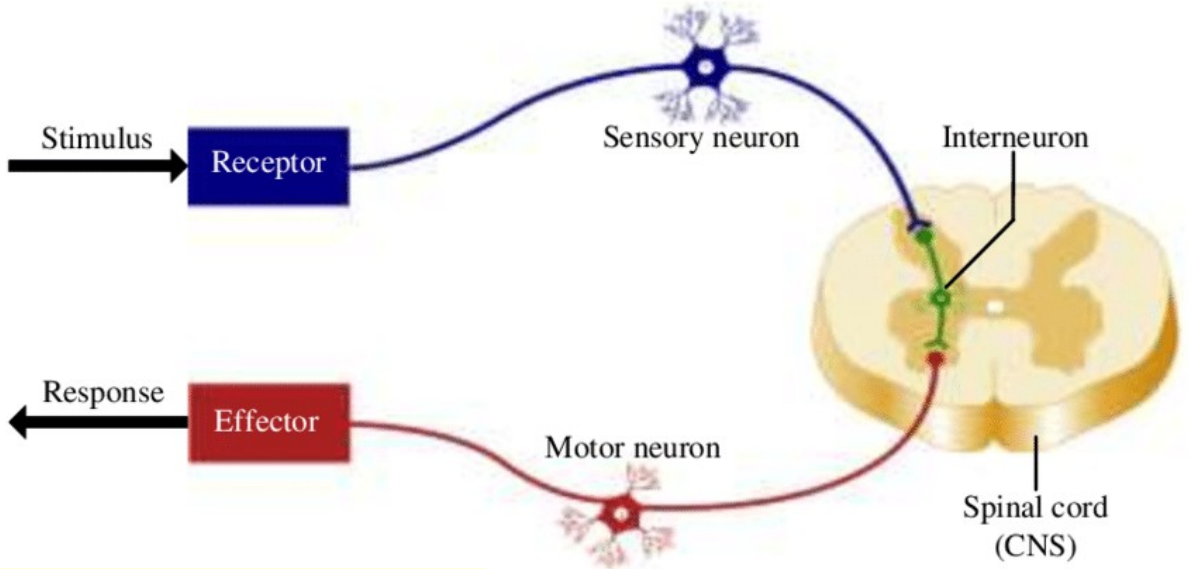
- Sensory neuron → interneurons → Motor Neuron
- Sensory neuron →Motor Neuron

A Motor neuron

- Carries impulse to effector

An Effector

- Carries out response.
- Muscle or secretory cells.



Learning Aim: describe the components of a reflex arc

Innate vs Acquired Reflexes

- Innate reflexes
  - Present from birth and/or appear during development
  - Automatic, not learned
  - Eg: blink reflex, patellar reflex

- Acquired reflexes
  - Complex motor patterns
  - Learned through repetition
  - Eg: balancing on a bike, jamming on brakes, catching a ball.

Learning Aim: explain the difference between innate and acquired reflexes

Some common reflexes

- Blink reflex – when someone pokes your eye
- Patellar reflex – knee jerk when hit with hammer
- Saliva secretion after smelling food
- Sneezing and coughing due to irritation
- Autonomic nervous system responses (eg fight/flight)
- Constriction of pupil after exposure to light

Learning Aim: list some common reflexes